

Aligning a Frozen Transformer with an External Memory

Experiment report — DecisionChains / memory_experiment

1 What we did

A pretrained **12-layer, 512-dim GPT-NeoX** was trained to execute *decision chains*: vectors $v_0 \in \mathbb{Z}_{10}^6$ generate 3–5 blocks, at each step one of two decision functions f, g picks the next letter-op. The two functions are:

$$f(v) = \begin{cases} 10 + v_1 & v_0 \text{ even} \\ v_1 & \text{else} \end{cases} \quad g(v) = \begin{cases} 10 + v_4 & v_3 \text{ even} \\ v_4 & \text{else} \end{cases}$$

Pretraining uses a fair coin flip between f and g . The resulting model picks each one $\approx 50\%$ of the time per step.

Goal: steer the model to always pick f without modifying the base model.

Memory module. One attention head with trainable keys $K \in \mathbb{R}^{N \times d}$, values $V \in \mathbb{R}^{N \times d}$, query projection $W_q \in \mathbb{R}^{d \times d}$, and optional scalar gate γ :

$$h' = h + \gamma \cdot \text{softmax}(W_q h \cdot K^\top / \sqrt{d}) V$$

Hooked at layer L ; base model frozen. At $N=32, d=512$: 294K trainable parameters ($\approx 1\%$ of the base model).

Training. Full-sequence CE on 3,000 mixed traces, with labels overridden at decision points to $f(v)$'s letter. AdamW, lr 3×10^{-3} , batch size 4, 10 epochs. ~ 3 minutes on one A100.

Finetuning baselines. Full FT (37.9M params) and LoRA rank-8 (294K params) in two supervision modes: **A** (ffff-only data, GT labels — standard LM loss) and **B** (all data with f -override at decision points — same signal the memory uses).

2 Evaluation

Two validation sets, both disjoint from training on both letter tuples and input vectors:

- **val_ffff** — 1,998 all- f examples, chain lengths 3–5 uniform.
- **val_full** — 2,000 mixed-pattern examples. The *honest* alignment test: the model encounters hidden states from coin-flip histories the memory may not have trained on.

Metrics (all evaluated via greedy generation):

- **f-selection (%)** — fraction of decision-point *steps* where the model picked f 's letter. Per-step average, so longer chains contribute more. Random baseline $\approx 52.5\%$.
- **Full alignment (%)** — fraction of *examples* where *every* step picked f 's letter. Strict conjunction: one wrong step scores zero. Baseline $\approx 8.8\%$ on mixed val (expected 0.525^n over chain lengths 3–5).
- **Op. accuracy (%)** — fraction of blocks with correct arithmetic (we re-execute each chosen letter's transformation and compare block text). Must not regress. Baseline $\approx 94.8\%$.

3 Main results

Table 1 compares the five memory variants and four FT baselines at matched compute ($n_{\text{train}}=3000$, epochs=10, batch size 4). Memory runs use 3 seeds; FT runs use 1.

The memory at layers 1 or 2 with $N=32$ matches or beats every FT variant on alignment while preserving arithmetic $\sim 4\times$ **better** than the closest LoRA baseline (-1.0 pp vs. -5.5 pp). A single memory entry ($N=1$, 263K parameters) is the *only* aligning method that *improves* arithmetic ($+1.5$ pp). Full FT mode B with 37.9M trainable parameters exhibits catastrophic forgetting (-10 pp op. accuracy).

Table 1: Main comparison, full-val (2k mixed-pattern examples). All numbers in %. Subscripts: change vs. pretrained baseline, in percentage points. *ns* = number of seeds. *Full align.* is the strict per-example metric (every step must be *f*).

Method	Params	Time	f-sel (%)	Full align (%)	Op. acc (%)
Pretrained (no training)	–	–	52.5	8.8	94.8
Memory , L=2 N=32 (3s)	294K	225s	97.3 _{+44.9}	88.0 _{+79.2}	93.3 _{−1.5}
Memory , L=1 N=32 (3s)	294K	177s	97.2 _{+44.8}	88.0 _{+79.1}	93.8 _{−1.0}
Memory , L=0 N=32 (3s)	294K	228s	96.1 _{+43.6}	82.5 _{+73.6}	91.8 _{−3.0}
Memory , L=2 N=1 (3s)	263K	208s	94.8 _{+42.3}	76.6 _{+67.7}	96.3 _{+1.5}
Memory , L=10 N=64 (3s)	327K	114s	65.8 _{+13.3}	16.3 _{+7.5}	90.5 _{−4.3}
FT LoRA, mode B	294K	252s	96.6 _{+44.1}	83.5 _{+74.6}	89.3 _{−5.5}
FT full, mode A	37.9M	272s	96.0 _{+43.5}	84.2 _{+75.3}	92.6 _{−2.1}
FT full, mode B	37.9M	360s	92.9 _{+40.5}	69.8 _{+60.9}	84.7 _{−10.0}
FT LoRA, mode A	294K	246s	81.6 _{+29.1}	37.2 _{+28.4}	94.3 _{−0.5}

ffff val (in-distribution sanity check)

Table 2: ffff val (1998 all-*f* examples, disjoint from train). Here every aligning method saturates. **complete_solution** (strictest: all ops correct + every letter valid + final vec matches GT output) is the meaningful metric. Baseline ffff: f-sel 76.8, full align 64.8, op. acc 95.1, complete_sol 73.5 (all %).

Method	f-sel	Full align	Op. acc	Complete sol
Memory L=2 N=32 (3s)	99.8	99.4	99.8	96.2±0.2
Memory L=1 N=32 (3s)	99.9	99.9	99.8	96.8 ±0.6
Memory L=0 N=32 (3s)	99.9	99.7	99.9	96.8±0.1
Memory L=2 N=1 (3s)	97.5	92.1	97.9	88.9±0.1
Memory L=10 N=64 (3s)	82.3	68.6	94.8	73.1±1.0
FT LoRA, mode B	100.0	100.0	99.9	96.8
FT LoRA, mode A	99.3	98.2	99.8	97.7
FT full, mode A	99.9	99.7	99.7	95.8
FT full, mode B	99.7	99.4	99.4	96.0

On ffff val, every aligning method reaches 99%+ alignment (because that is what they were trained for). FT LoRA mode A is actually the single best on **complete_solution**, but it drops to 37% full alignment on mixed val (Table 1) — a gap of 61 pp that reveals the off-path generalisation limit of training on ffff data alone.

4 Ablations

4.1 Layer ablation

Table 3 fixes $n_{\text{train}}=3000$, $N=32$ and varies only the injection layer.

Layer 1 $\approx$ Layer 2 > Layer 0 $\gg$ Layer 10. Injecting into an early residual layer allows the correction to propagate through all downstream blocks. The $L=10$ configuration (3 seeds) shows high variance (± 6.5 pp on f-selection), suggesting the memory loses effectiveness when injected too close to the output.

4.2 Memory size $\times$ training size

Table 4 sweeps n_{train} (training examples) and N (memory entries) at $L=2$. All single-seed, $n_{\text{eval}}=200$.

Table 3: Injection layer. Layers 0–2 evaluated on full 2k val with 3 seeds; layers 4–11 evaluated on earlier Phase-1 sweep (50 examples, 1 seed) — numbers are rougher but trend is clear. Layer 2 is optimal; later layers degrade sharply.

Layer	f-sel Δ (pp)	Full align Δ (pp)	Op. acc Δ (pp)	Eval
0	+43.6 $\pm$ 0.5	+73.6 $\pm$ 1.7	−3.0 $\pm$ 0.9	2k, 3 seeds
1	+44.8 $\pm$ 0.2	+79.1 $\pm$ 1.1	− 1.0 $\pm$ 0.9	2k, 3 seeds
2	+ 44.9 $\pm$ 0.3	+ 79.2 $\pm$ 1.3	−1.5 $\pm$ 0.1	2k, 3 seeds
4	+38.7	+58.0	−1.1	50, 1 seed
6	+36.7	+52.0	−5.2	50, 1 seed
8	+26.1	+28.0	−1.9	50, 1 seed
10	+17.4 $\pm$ 0.4	+11.3 $\pm$ 5.0	−2.8 $\pm$ 4.8	50, 3 seeds
11	+20.5	+16.0	−4.8	50, 1 seed

Table 4: Full alignment (%) at $L=2$ across n_{train} (rows) and N (columns). Baseline is 8%. Bold: row best. Shaded cells $\geq 80\%$.

n_{train}	$N=1$	$N=2$	$N=4$	$N=8$	$N=16$	$N=32$	$N=64$
100	12.5	12.5	17.0	15.5	26.0	24.0	17.5
1000	55.5	60.5	80.0	80.5	74.5	79.5	78.0
2000	67.0	86.0	83.5	82.5	90.5	89.5	80.5
3000	73.2 $\pm$ 2.4	74.0	89.0	89.5	87.0	88.0 $\pm$ 2.6	89.5

Key pattern. The memory saturates early in N : by $N=4$ –8 we capture most of the gain; by $N=16$ –32 the ceiling is reached. Training data is the harder bottleneck: $n=100$ plateaus near 26% full alignment; $n \geq 1000$ crosses 80%; $n=3000$ saturates at $\sim 91\%$.

4.3 Where is the knee? (in-distribution view)

Table 5 fixes $L=2$, $N=32$ and varies n_{train} , reporting all four in-distribution (ffff val) metrics. This shows where training data starts paying off.

Table 5: ffff val metrics at $L=2$, $N=32$ across n_{train} . `complete_solution` (strictest: correct arithmetic + valid letters + final vec matches GT output) is the honest in-distribution metric. Baseline: f-sel 72.9, full align 59.0, op. acc 94.8, complete 72.0 (all %, $n_{\text{eval}}=200$).

n_{train}	f-sel	Full align	Op. acc	Complete sol
100	86.2	64.5	96.7	69.0
1000	97.6	93.0	99.2	90.5
2000	99.3	97.5	98.8	94.0
3000	99.7 $\pm$ 0.3	99.3 $\pm$ 0.6	99.7 $\pm$ 0.3	96.7$\pm$0.6

Reading the knee. At $n=100$, the memory under-trains: f-sel crosses baseline (+13 pp) but `complete_solution` actually drops below baseline (−3 pp) — the small correction causes more arithmetic errors than it prevents letter errors. Between $n=100$ and $n=1000$ all metrics jump ~ 20 pp. From $n=1000$ onward gains are logarithmic: +4 pp on `complete_solution` going to $n=2000$, then +2.7 pp to $n=3000$. **The practical sweet-spot sits around $n=1000$ –2000**; beyond that the in-distribution ceiling is already effectively reached.

4.4 Ingredient ablation

Table 6 ablates the recipe at $L=10$, $N=32$ (Phase-1 sweep, $n_{\text{eval}}=50$). Each row changes one dimension from the reference.

Order of importance: (1) the f -override at decision points is the actual alignment signal — without it, mixed training does nothing ($A5$, $\Delta \approx 0$); (2) mixing the training distribution matters — ffff-only training

Table 6: What breaks the recipe. Layer 10 reference so differences are bigger. Dropping full-sequence CE destroys arithmetic (-46 pp). Dropping the f -override eliminates the alignment signal.

Config	Change from reference	f-sel Δ	Align Δ	Op Δ
A1_ref	(full recipe, 3 seeds)	$+17.4 \pm 0.4$	$+11.3 \pm 5.0$	-2.8 ± 4.8
A2_no_mix	ffff-only data, GT labels	$+3.0$	$+2.0$	$+0.1$
A3_no_fullseq	DP-only CE	-4.3	-2.0	-46.2
A4_no_gate	no learnable γ	$+11.5$	$+4.0$	-1.0
A5_no_override	GT labels, no f -override	-0.3	0.0	$+2.2$

generalises poorly to off-path hidden states (A2); (3) full-sequence CE is essential — DP-only CE breaks arithmetic catastrophically (A3); (4) the gate contributes a small, consistent boost (A4).

5 Takeaways

1. **A small attention-head-shaped module, attached at an early residual layer, matches or beats full finetuning on alignment while preserving arithmetic roughly $4\times$ better.** The gain is architectural, not a consequence of matching supervision: LoRA at the same parameter count with the same f -override signal loses by ~ 4 pp on op. accuracy.
2. **$N=1$ is sufficient for high alignment and *improves* arithmetic.** Combined with the linear probe findings earlier in the project, this is consistent with the hypothesis that the f/g preference is controlled by (approximately) a single direction in layer-2 residual space.
3. **Full FT with mode B breaks.** With 37.9M free parameters, the override-on-mixed-data signal causes catastrophic forgetting (-10 pp op. accuracy). LoRA on the same supervision stays stable because the adapter can only move the model in a low-rank subspace.
4. **Early layers dominate.** $L \in \{1, 2\}$ are tied optimal. $L = 10$ (high variance across seeds) barely works. This suggests that for inference-time alignment of this kind, the correction should be injected where the downstream network has most room to propagate it.